

Lecture 5: Maximum Likelihood

Probabilistic Machine Learning

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One example will build a general method

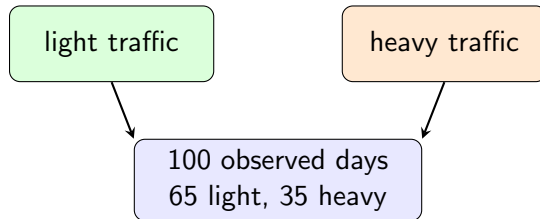
- 1 Begin with one unknown probability and observed traffic data.
- 2 Introduce a model, likelihood, log-likelihood, and MLE only when the example needs them.
- 3 Extract a procedure that applies to other models and data types.
- 4 Test it on Gaussian travel times and next-token counts, then connect maximum likelihood to empirical distributions and KL divergence.

Question for today

How can observed data tell us which distribution parameter to use?

A city knows the data but not the probability

A road has either **light** or **heavy** traffic each day.



Let θ^* be the unknown probability of light traffic.

The estimation problem

What value should the city use for θ^* : 0.2, 0.4, 0.6, 0.8, or something else?

A probabilistic model makes the question precise

Define one random variable for each day:

$$X_i = \begin{cases} 1, & \text{light traffic,} \\ 0, & \text{heavy traffic.} \end{cases}$$

A proposed parameter value $\theta \in [0, 1]$ specifies

$$X_i \sim \text{Bernoulli}(\theta), \quad p(x_i; \theta) = \theta^{x_i} (1 - \theta)^{1-x_i}.$$

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Three objects to keep separate

X_i	x_i	θ
random variable	observed value	proposed parameter

Independence turns 100 observations into one product

Let the full dataset be $\mathcal{D} = \mathbf{x} = (x_1, \dots, x_{100})$.

If the days are independent and have the same parameter,

$$\begin{aligned} p(\mathbf{x}; \theta) &= \prod_{i=1}^{100} p(x_i; \theta) \\ &= \prod_{i=1}^{100} \theta^{x_i} (1 - \theta)^{1-x_i} \\ &= \theta^{65} (1 - \theta)^{35}. \end{aligned}$$

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Modeling assumption

The product is justified by the **iid assumption**. If traffic patterns change over time or days influence one another, this model may need to change.

The same expression has two different jobs

Probability model

Fix θ and vary possible data $\mathbf{x}$:

$$p(\mathbf{x}; \theta).$$

It answers: *What data might this model produce?*

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Likelihood function

Fix the observed data and vary θ :

$$L(\theta) = p(\mathbf{x}; \theta).$$

It answers: *Which parameter values explain these data well?*

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It answers: *Which parameter values explain these data well?*

A likelihood is a function of θ ; it is not generally a probability distribution over θ .

Candidate values make the idea visible

If we record only the count $K = 65$, then

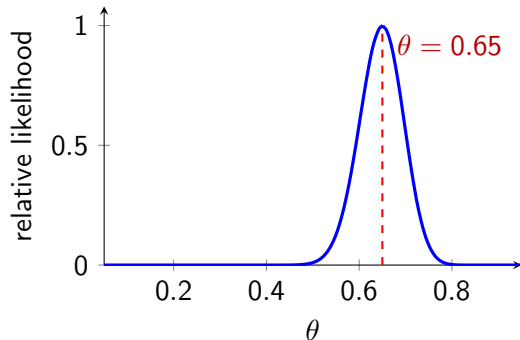
$$L(\theta) = \binom{100}{65} \theta^{65} (1 - \theta)^{35}.$$

Candidate θ	$p(K = 65; \theta)$
0.2	1.6×10^{-22}
0.4	2.6×10^{-7}
0.6	0.0491
0.8	0.00019

Likelihood principle

Prefer a parameter that assigns greater probability to the observed data.

The whole curve turns comparison into optimization



We are not limited to four candidates:

$$\hat{\theta}_{\text{mle}} = \arg \max_{\theta \in [0,1]} L(\theta)$$

The data are fixed. We search across the permitted parameter values.

Logs preserve the maximizer and simplify the algebra

Because log is strictly increasing,

$$\arg \max_{\theta} L(\theta) = \arg \max_{\theta} \log L(\theta).$$

For the traffic data,

$$\begin{aligned}\ell(\theta) &= \log L(\theta) \\ &= \log \binom{100}{65} + 65 \log \theta + 35 \log(1 - \theta) \\ &\doteq 65 \log \theta + 35 \log(1 - \theta).\end{aligned}$$

Two reusable simplifications

Products become sums, and terms constant in θ can be dropped without changing the MLE. The symbol $\doteq$ marks equality up to an additive constant independent of θ .

The sample fraction solves the traffic problem

Differentiate the log-likelihood:

$$\frac{d\ell}{d\theta} = \frac{65}{\theta} - \frac{35}{1-\theta}.$$

Set it equal to zero:

$$\frac{65}{\theta} - \frac{35}{1-\theta} = 0 \iff 65(1-\theta) = 35\theta$$

$$\iff \hat{\theta}_{\text{mle}} = \frac{65}{100} = 0.65.$$

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Interpretation

Under the iid Bernoulli model, maximum likelihood chooses the observed fraction of light-traffic days.

MLE is a reusable five-step procedure

Step	Question to answer
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- | | |
|---|---|
| 1 | What probabilistic model and parameter space are assumed? |
| 2 | What is the joint probability of the observed data? |
| 3 | Which factors or terms are constant in the parameter? |
| 4 | What parameter maximizes the likelihood or log-likelihood? |
| 5 | Is the answer feasible, a maximum, and sensible in context? |
-

model $\longrightarrow L(\theta) \longrightarrow \ell(\theta) \longrightarrow \hat{\theta}_{\text{mle}} \longrightarrow \text{check.}$

The Bernoulli derivation works for any n and k

Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ and the data contain k successes.

$$L(\theta) = \theta^k (1 - \theta)^{n-k},$$

$$\ell(\theta) = k \log \theta + (n - k) \log(1 - \theta),$$

$$\ell'(\theta) = \frac{k}{\theta} - \frac{n - k}{1 - \theta}.$$

Solving $\ell'(\theta) = 0$ gives

$$\hat{\theta}_{\text{mle}} = \frac{k}{n}.$$

A second frequentist route agrees here

The numbers 65 and 100 were not special. Moment matching sets $\mathbb{E}_{\theta}[X] = \theta$ equal to the sample mean k/n , giving the same estimate for this Bernoulli model.

For a continuous parameter, optimization becomes calculus

Let $\theta \in \Theta \subseteq \mathbb{R}$ and suppose $\ell(\theta)$ is differentiable. For an *interior* maximizer,

$$\underbrace{\ell'(\hat{\theta})}_{\text{score}} = 0.$$

A stationary point is only a candidate. We must still check

$$\ell''(\hat{\theta}) < 0 \quad \text{or compare likelihood values and boundaries.}$$

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For a vector parameter $\boldsymbol{\theta} \in \mathbb{R}^d$, the same idea becomes

$$\nabla \ell(\hat{\boldsymbol{\theta}}) = \mathbf{0}, \quad \nabla^2 \ell(\hat{\boldsymbol{\theta}}) \text{ negative definite at a strict local maximum.}$$

Do not forget the parameter space

A maximum may occur on the boundary, where the score need not be zero.

Daily variation determines the travel-time distribution

The city records the travel time on the same route for n days. Model day i as

$$Y_i = \mu^* + Z_i, \quad Z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, (\sigma^*)^2).$$

Here μ^* is the route's typical travel time and Z_i is day-to-day variation. Adding μ^* shifts the mean but not the variance, so

$$Y_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu^*, (\sigma^*)^2).$$

After n days, the random vector $\mathbf{Y} = (Y_1, \dots, Y_n)$ becomes the recorded travel times $\mathbf{y} = (y_1, \dots, y_n)$.

The chapter considers two cases

- (1) σ^* known and μ^* unknown; (2) both μ^* and σ^* unknown.

Start with the density of one day's travel time

First suppose the day-to-day standard deviation σ is known. For a candidate mean travel time μ , the model assigns density

$$p_{Y_i}(y; \mu) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{y - \mu}{\sigma}\right)^2\right].$$

Evaluating this density at the recorded travel time y_i gives

$$p_{Y_i}(y_i; \mu) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{y_i - \mu}{\sigma}\right)^2\right].$$

What changes after recording the day?

The travel time y_i is fixed; the candidate mean μ varies.

Independence turns daily densities into a likelihood

Because the daily travel times are independent, their joint density factors:

$$\begin{aligned} L(\mu) &= p_{\mathbf{Y}}(\mathbf{y}; \mu) \\ &= \prod_{i=1}^n p_{Y_i}(y_i; \mu) \\ &= \prod_{i=1}^n \left\{ \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{y_i - \mu}{\sigma} \right)^2 \right] \right\} \\ &= (\sigma\sqrt{2\pi})^{-n} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right]. \end{aligned}$$

Why this is a likelihood

The recorded times $\mathbf{y}$ are fixed, and $L(\mu)$ is viewed as a function of the candidate mean travel time μ .

Taking logs exposes the optimization problem

Apply $\log \prod_i a_i = \sum_i \log a_i$:

$$\begin{aligned}\ell(\mu) &= \log L(\mu) \\ &= \sum_{i=1}^n \log p_{Y_i}(y_i; \mu) \\ &= \sum_{i=1}^n \left[-\log(\sigma\sqrt{2\pi}) - \frac{1}{2} \left(\frac{y_i - \mu}{\sigma} \right)^2 \right] \\ &= -n \log(\sigma\sqrt{2\pi}) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \\ &\doteq -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2.\end{aligned}$$

The dropped term is constant in μ because σ is known.

Known variability: the score gives the mean travel time

Differentiate the log-likelihood term by term:

$$\begin{aligned}\frac{d\ell}{d\mu} &= -\frac{1}{2\sigma^2} \sum_{i=1}^n 2(y_i - \mu)(-1) \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \mu).\end{aligned}$$

Set the score equal to zero and solve:

$$\begin{aligned}\frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \mu) = 0 &\iff \sum_{i=1}^n y_i - n\mu = 0 \\ &\iff \hat{\mu}_{\text{mle}} = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}.\end{aligned}$$

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Since $\ell''(\mu) = -n/\sigma^2 < 0$, this stationary point is the unique maximum.

Unknown variability adds one new score equation

Now treat both $\mu \in \mathbb{R}$ and $\sigma > 0$ as unknown:

$$\ell(\mu, \sigma) \doteq -n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2.$$

The derivative with respect to μ is unchanged from the known- σ case:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \mu).$$

What changed?

The parameter is now (μ, σ) . We retain the mean score equation and add a second score equation for the unknown variability.

Then differentiate with respect to travel-time variability

Write the terms involving σ using powers:

$$\ell(\mu, \sigma) \doteq -n \log \sigma - \frac{1}{2} \sigma^{-2} \sum_{i=1}^n (y_i - \mu)^2.$$

Differentiating term by term gives

$$\begin{aligned} \frac{\partial \ell}{\partial \sigma} &= -\frac{n}{\sigma} - \frac{1}{2} \sum_{i=1}^n (y_i - \mu)^2 (-2\sigma^{-3}) \\ &= -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (y_i - \mu)^2. \end{aligned}$$

The chain-rule step for σ

Since $\frac{d}{d\sigma} \sigma^{-2} = -2\sigma^{-3}$, differentiating the squared-error term produces the positive σ^{-3} term.

Solving the score equations gives both estimates

Set the mean score equal to zero:

$$\frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \mu) = 0 \iff \sum_{i=1}^n y_i - n\mu = 0 \iff \boxed{\hat{\mu}_{\text{mle}} = \bar{y}.}$$

Substitute $\hat{\mu} = \bar{y}$ into the σ score equation:

$$-\frac{n}{\hat{\sigma}} + \frac{1}{\hat{\sigma}^3} \sum_{i=1}^n (y_i - \bar{y})^2 = 0 \iff -n\hat{\sigma}^2 + \sum_{i=1}^n (y_i - \bar{y})^2 = 0$$
$$\iff \boxed{\hat{\sigma}_{\text{mle}}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2.}$$

The MLE uses n , not $n - 1$

The familiar $n - 1$ denominator is a later bias correction, not the MLE.

The curvature check confirms a maximum

At $\hat{\mu} = \bar{y}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_i (y_i - \bar{y})^2$, the mixed derivative is zero because $\sum_i (y_i - \bar{y}) = 0$.

The diagonal Hessian entries are

$$\begin{aligned}\left. \frac{\partial^2 \ell}{\partial \mu^2} \right|_{(\hat{\mu}, \hat{\sigma})} &= -\frac{n}{\hat{\sigma}^2} < 0, \\ \left. \frac{\partial^2 \ell}{\partial \sigma^2} \right|_{(\hat{\mu}, \hat{\sigma})} &= \frac{n}{\hat{\sigma}^2} - \frac{3}{\hat{\sigma}^4} \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= -\frac{2n}{\hat{\sigma}^2} < 0.\end{aligned}$$

Hence the Hessian is negative definite and the stationary point is a strict local maximum, provided $\hat{\sigma} > 0$.

The recipe stays fixed; the ingredients change

	Bernoulli traffic label	Gaussian travel time
Parameter	$\theta \in [0, 1]$	$\mu \in \mathbb{R}$
Data summary	successes k	sample sum $\sum_i y_i$
Log-likelihood	$k \log \theta + (n - k) \log(1 - \theta)$	$-\frac{1}{2\sigma^2} \sum_i (y_i - \mu)^2$
MLE	k/n	$\bar{y}$

General lesson

There is no universal MLE formula. There is a universal procedure whose details are determined by the assumed sampling model.

A valid MLE requires more than taking a derivative

After solving a first-order equation, check:

- **Parameter space:** Is the answer allowed? For Bernoulli, $0 \leq \theta \leq 1$.
- **Boundaries:** If all observations are failures or successes, the MLE is at 0 or 1.
- **Maximum:** Compare candidates, inspect the likelihood, or check curvature.
- **Model assumptions:** Were the observations plausibly iid?
- **Interpretation:** Does the estimate have the right units and a sensible relationship to the data?

Discrete parameter spaces

If Θ is finite, no derivative is needed: evaluate $L(\theta)$ for every permitted candidate and choose the largest.

Normalized counts are an MLE and an empirical distribution

For one text context c , let token w_j appear N_j times among n observed continuations. Its empirical probability is $\hat{p}_j = N_j/n$.

A categorical model proposes probabilities $\mathbf{q} = (q_1, \dots, q_m)$:

$$L(\mathbf{q}) = \prod_{j=1}^m q_j^{N_j}, \quad q_j \geq 0, \quad \sum_{j=1}^m q_j = 1.$$

Maximizing this constrained likelihood gives

$$\hat{q}_j = \frac{N_j}{n} = \hat{p}_j.$$

Fit, then plug in

With counts (5, 3, 2) for (sat, slept, EOS), both the empirical distribution and the fitted predictive distribution are (0.5, 0.3, 0.2).

KL divergence measures a model's mismatch with the data

Compare an empirical distribution $\hat{\mathbf{p}}$ with a proposed model distribution $\mathbf{q}$ using

$$D_{\text{KL}}(\hat{\mathbf{p}} \parallel \mathbf{q}) = \sum_{j=1}^m \hat{p}_j \log \frac{\hat{p}_j}{q_j}.$$

For the next-token counts (5, 3, 2),

$$\hat{\mathbf{p}} = (0.5, 0.3, 0.2).$$

Candidate model	$\mathbf{q}$	$D_{\text{KL}}(\hat{\mathbf{p}} \parallel \mathbf{q})$
A	(0.5, 0.3, 0.2)	0
B	(0.4, 0.4, 0.2)	≈ 0.025

Smaller KL divergence means the model distorts the observed relative frequencies less. It is always nonnegative and is zero only when the two distributions agree.

Maximizing likelihood minimizes forward KL divergence

Because $N_j = n\hat{p}_j$, the categorical log-likelihood is

$$\ell(\mathbf{q}) = \sum_{j=1}^m N_j \log q_j = n \sum_{j=1}^m \hat{p}_j \log q_j.$$

Meanwhile,

$$D_{\text{KL}}(\hat{\mathbf{p}}\|\mathbf{q}) = \sum_j \hat{p}_j \log \hat{p}_j - \sum_j \hat{p}_j \log q_j.$$

The first term is fixed by the data. For any model family $\mathbf{q} = \mathbf{p}_\theta$, therefore,

$$\hat{\theta}_{\text{mle}} = \arg \min_{\theta} D_{\text{KL}}(\hat{\mathbf{p}}\|\mathbf{p}_\theta).$$

Direction matters

MLE uses forward KL: the empirical distribution is the first argument.

A fitted distribution becomes a plug-in predictor

After fitting $\hat{\theta}$, use $p_{\hat{\theta}}$ as the working distribution for prediction.

Traffic

With $\hat{\theta} = 0.65$,

$$\Pr(\text{light tomorrow}) \approx 0.65.$$

Next token

Use $\hat{\mathbf{q}} = (0.5, 0.3, 0.2)$ to score or sample (sat, slept, EOS).

A general supervised model

From the fitted conditional $p_{\hat{\theta}}(y | x)$:

$$\hat{f}_{0-1}(x) = \arg \max_y p_{\hat{\theta}}(y | x),$$

$$\hat{f}_{\text{sq}}(x) = \mathbb{E}_{\hat{\theta}}[Y | X = x].$$

Two stages

Estimation fits a distribution; plug-in prediction acts as though that fitted distribution were the truth.

Two interpretations lead to one unresolved question

Maximum likelihood can be described in two equivalent ways:

- choose the parameter that makes the observed data most likely;
- choose the allowed distribution closest to the empirical distribution in forward KL divergence.

Both describe fit to this observed sample. Neither tells us:

- how much the estimate would change under a different sample;
- whether the estimator is biased or highly variable;
- how much data are needed for reliable estimation; or
- whether the model family is a good description of reality.

Next lecture

Treat the estimator as a random variable and study bias, variance, mean squared error, and consistency.

What should stay with you

- ① A likelihood fixes the observed data and varies the parameter.
- ② iid observations give a product likelihood; logs turn it into a sum.
- ③ MLE selects the allowed parameter with the greatest likelihood.
- ④ The sampling model determines the algebra and the resulting MLE.
- ⑤ MLE minimizes forward KL divergence; the fitted distribution can then drive plug-in predictions.
- ⑥ Always check the parameter space, boundaries, maximum, assumptions, and interpretation.

model $\rightarrow L \rightarrow \ell \rightarrow \text{optimize} \rightarrow \text{check}$

Group activity: apply and extend the method

Work in groups of two or three. Submit one response per group.

- 1 **Bernoulli warm-up:** For 8 light-traffic days out of 12, derive the unrestricted MLE of θ .
- 2 **Finite candidates:** If the city permits only $\Theta = \{0.4, 0.6, 0.8\}$, compare log-likelihoods and select the MLE.
- 3 **Categorical fit:** A context is followed by (sat, slept, EOS) with counts (5, 3, 2). Maximize its categorical likelihood subject to the probabilities summing to one. A Lagrange multiplier may help.
- 4 **KL interpretation:** Compare your fitted distribution with $\mathbf{q} = (0.4, 0.4, 0.2)$ using $D_{\text{KL}}(\hat{\mathbf{p}}\|\mathbf{q})$. Explain why maximum likelihood selects the distribution with the smaller value.